

MolmoAct2 · arXiv 2605.02881 · Ai2 / allenai

Archaeologist

11+3 · all seats empty

Parallel Molmo vs PI lineages

FAST→FM hybrid · Think $\neq$ HL $\neq$ stages

Oct 28 paper club · VLA Architectures · quiet build

Spine: geometric S2 $\neq$ semantic HL $\neq$ stage voting · FM motor

FALSE: “has $\pi_{0.5}$ HL” / “has RTC” / “has BEHAVIOR S2 voting”

CRITICAL SPINE — three outer loops (do NOT conflate)

All three call an FM chunk expert (System 1 / motor). Different intermediate discrete states.

MolmoAct2 Think

Adaptive depth / geometry

- 10×10 depth codes (100)
- VQ depth lattice
- Regen if RGB cos < 0.996
- Cached static cells
- Geometric System-2

NOT semantic subtasks

$\pi_{0.5}$ HL

Semantic subtask language

- AR subtask strings
- Optional boxes
- Task language hierarchy
- Then FM LL chunk
- Semantic System-2

NOT depth lattice

BEHAVIOR'25 S2

Stage IDs + voting

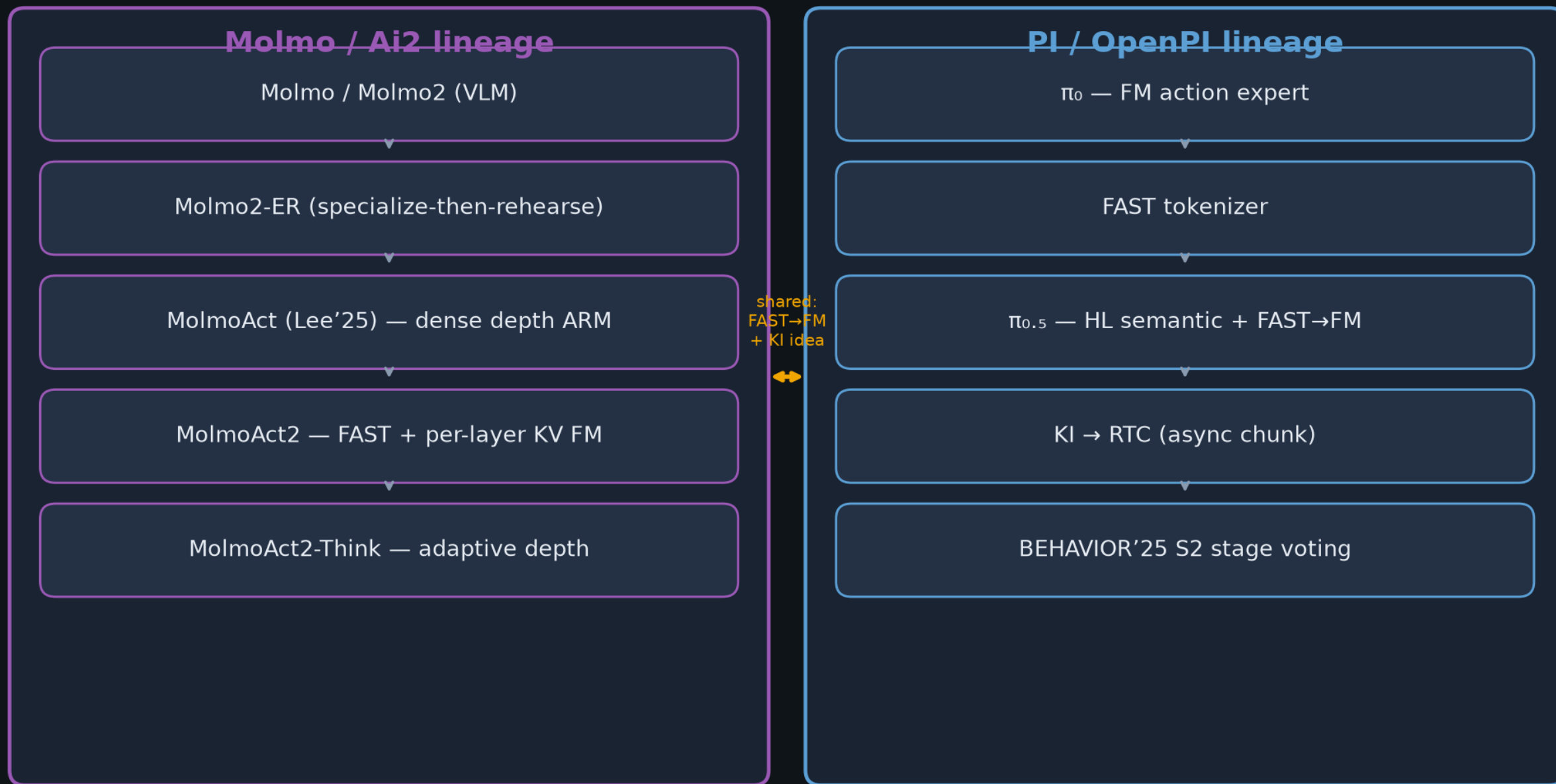
- Task-progress stages
- Classifier + voting
- Non-Markov memory
- Then FM expert
- Checklist System-2

NOT scene geometry

FALSE claims: “has $\pi_{0.5}$ HL” · “has RTC” · “has BEHAVIOR S2 voting”

TENTATIVE: BEHAVIOR may want depth + stage complementary · RTC absent — latency via FM cache + CUDA Graphs + Think reuse

Parallel lineages — Molmo ARM vs PI stack



What's new vs reused (Archaeology)

NEW

- ◆ Molmo2-ER recipe + 3.3M ER data
- ◆ 720 h BimanualYAM open set
- ◆ Open FAST mixture (5 emb.)
- ◆ Per-layer KV (vs final hidden)
- ◆ Adaptive depth temporal reuse
- ◆ Large open multi-emb OOTB study

REUSED

- Molmo2 visual interface
- MolmoAct depth VQ idea
- FAST tokenization principle
- FM / DiT action experts
- KI stop-grad idea (post)
- Chunked BC/FM control family

Citation hygiene — FALSE / CLEAR / TENTATIVE

FALSE

“MolmoAct2 has $\pi_{0.5}$ HL subtasks”

Uses depth lattice, not semantic HL

FALSE

“Implements RTC”

RTC absent; cache + Graphs + Think reuse

FALSE

“Has BEHAVIOR System-2 voting”

Different intermediate (geometry $\neq$ stages)

CLEAR

Hybrid FAST $\rightarrow$ FM curriculum like $\pi_{0.5}$

Structural analogy; shared KI idea

CLEAR

Both call FM chunk expert (S1)

Different L1 tokens into same motor family

TENTATIVE

RTC wrap on Molmo FM

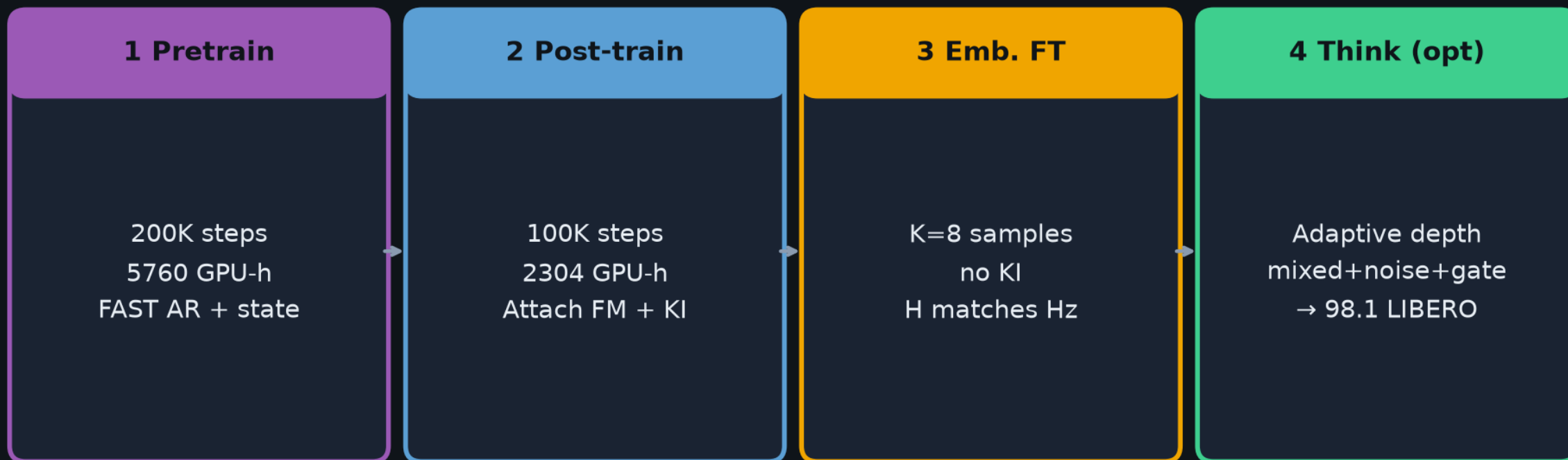
High-leverage follow-on — not in PDF

TENTATIVE

Depth + stage complementary

Likely for BEHAVIOR long-horizon

Curriculum — FAST discrete → FM continuous



Deploy default = continuous FM path (not discrete AR actions)

Expert $\sim 621\text{M}$ · L=36 · AdaRMS on t · N=10 Euler · $H \in \{10, 15, 30\}$ · $D \leq 32$

L0

Motion / Perception + language **— L0 → L1 → L2**

SigLIP2 + LLM multi-cam

Interface: tokens + per-layer K/V → expert



L1

System-2 (optional Think)

AR depth codes (10×10)

Interface: depth → VLM context → expert KV



L2

Motor

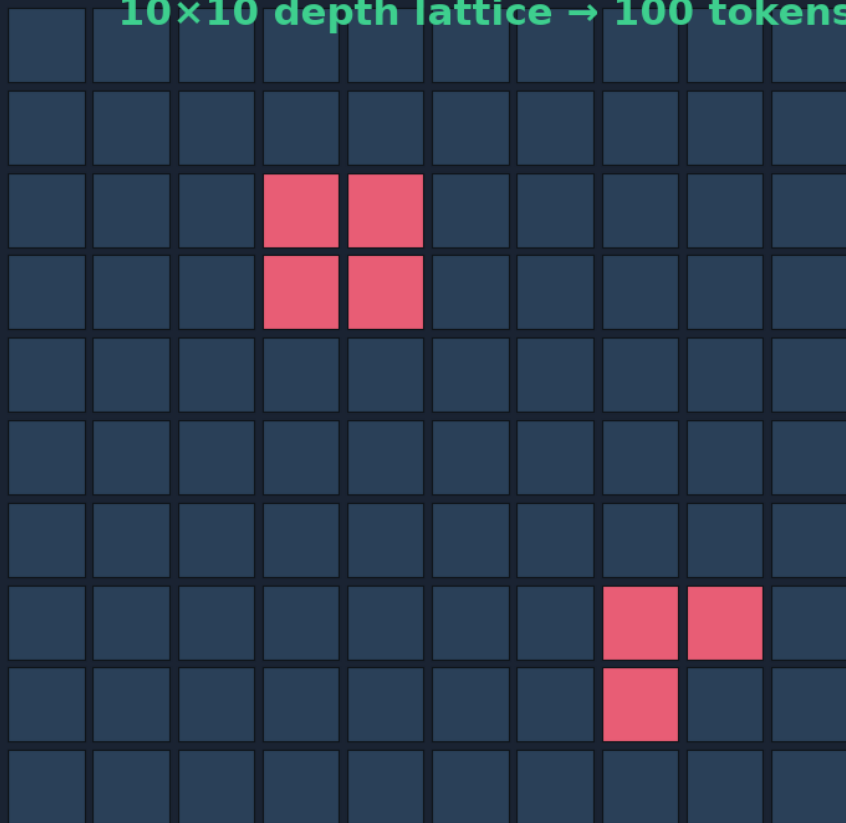
DiT FM · $H \approx H_z$ · N=10 Euler

Interface: continuous joints / delta-EE

MolmoAct2-Think — adaptive depth codes (geometric S2)

NOT $\pi_{0.5}$ semantic HL · NOT BEHAVIOR stage IDs

10×10 depth lattice → 100 tokens



pink = regenerate (RGB cos < 0.996) · blue-gray = cache reuse

1. Observe RGB (+ multi-cam)

2. AR depth codes <depth_0...127>

3. Reuse static cells / regen changed

4. Optional per-layer depth gate

5. Gate into expert KV → FM chunk

Latency tax: 12.7 Hz Think vs 55.8 Hz flat (H100, H=10, Graphs)

Fig.4 · Architecture: VLM + per-layer KV → FM expert

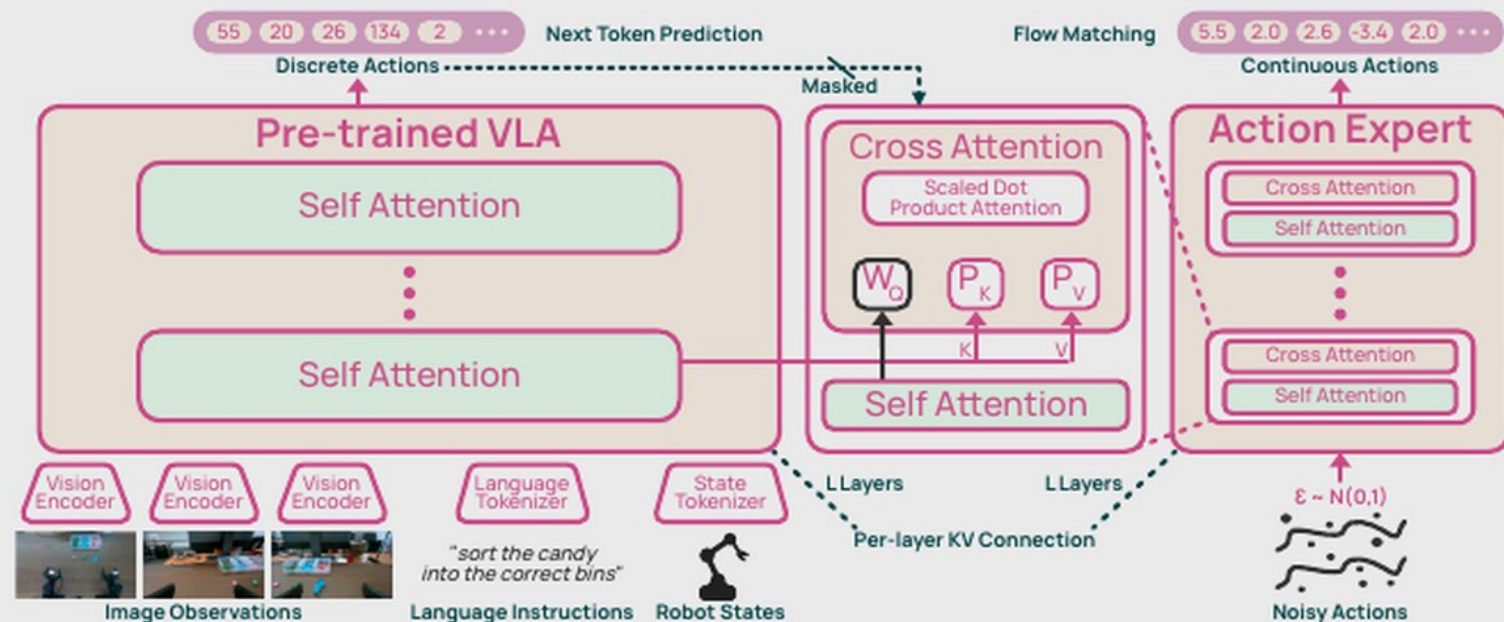


Figure 4 Overview of MolmoAct2. Image observations, language instructions, and robot states are tokenized and processed by a pre-trained VLA backbone with self-attention layers. Post-training attaches a DiT-style action expert with the same number of layers as the backbone. At each layer, the backbone key and value tensors are projected and reused as the key and value inputs to the action expert's cross-attention, enabling per-layer KV conditioning from visual-language context to continuous control. The expert is trained with flow matching to denoise a noisy action trajectory into a continuous robot trajectory. During training, the backbone is also supervised with next-token prediction over discrete action tokens, while the target action-token span is masked from the expert so continuous action prediction cannot condition on the ground-truth discrete actions.

control strings using the same special-token wrappers used during training, e.g., `<setup_start>bimanual yam robotic arms in molmoact2<setup_end>`, and `<control_start>absolute joint pose<control_end>` or `<control_start>delta end-effector pose<control_end>`; the corresponding chat template and output trigger formatting are detailed in Appendix B.

Action and state representation. This stage is deliberately limited to discrete autoregressive supervision: the model predicts action tokens, but does not yet train the continuous action expert used in post-training. Each

Fig.5 · MolmoAct2-Think adaptive depth

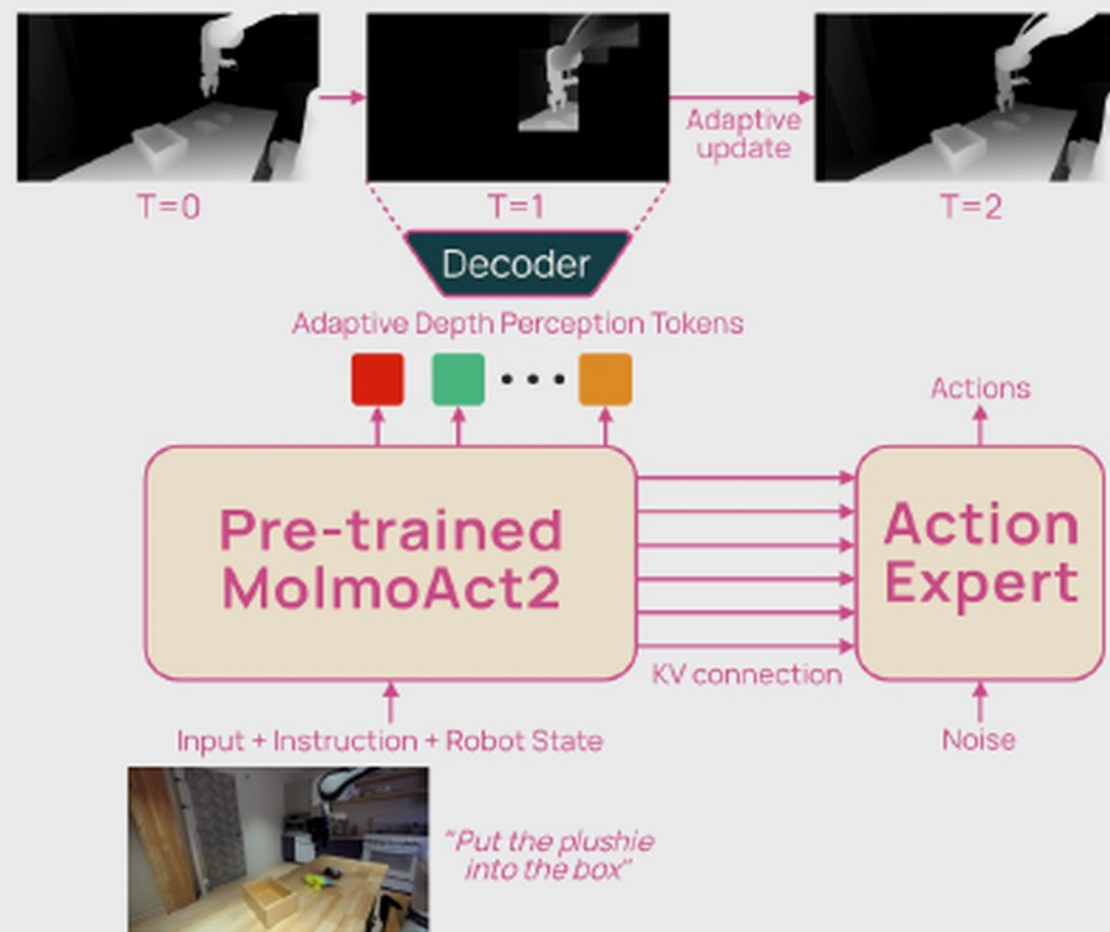


Figure 5 Overview of MolmoAct2-Think. MOLMOACT2-THINK augments the MOLMOACT2 action-generation pipeline with adaptive depth-token reasoning, reusing cached depth codes for static regions and regenerating depth codes for changed regions before conditioning the action expert.

We use annotated language instructions, absolute joint-pose actions, and a 30-step action chunk for the 30 Hz

Fig.1 · MolmoAct2 overview



Figure 1 Overview of MolmoAct2. MOLMOACT2 is a fully open action reasoning model for real-world deployment. From a suite of high-quality robot datasets that we collect, filter, and curate at scale across three platforms spanning the low-to-medium cost range (**left**), we train MOLMOACT2 and its adaptive-depth reasoning variant MOLMOACT2-THINK, coupled to the VLM backbone through per-layer KV conditioning (**center**). The resulting models deploy out-of-the-box on bimanual YAM, SO-100/101, and DROID Franka, and adapt to in-the-wild tasks such as cleaning up, washing dishes, wetlab automation, and pouring tea (**right**).

1 Introduction

Physical intelligence is fundamentally organized around perception and action (Tversky, 2025). Rather than reasoning over abstract internal computation, human think by constructing spatial representations, simulating actions, and interacting with the world through their bodies (Tversky, 2019, 2013). Although we conflate today's robot foundation models as displaying such intelligence; from a cognitive science perspective, they remain incomplete models of intelligence. They often lack structured spatial representations Qu et al. (2025), rely on heavyweight internal reasoning processes that impede real-time interaction (Lee et al., 2025; Kim et al., 2026a; Zhu et al., 2025), and are difficult to adapt or extend due to limited openness (Black et al., 2024; Intelligences et al., 2025).

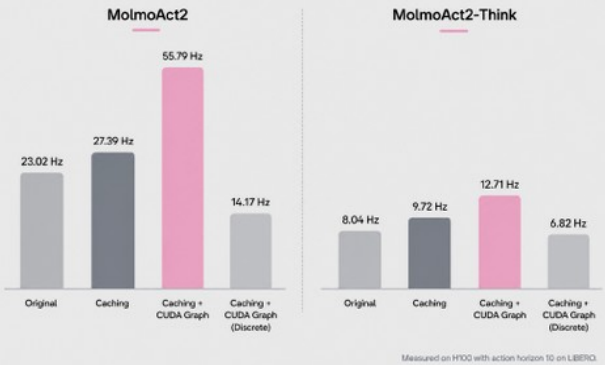


Figure 8 Inference control rate after caching and CUDA Graph optimizations. Control rate is action horizon divided by end-to-end action-generation latency; measurements use horizon 10 on LIBERO with a single H100.

MOLMOACT2 also exposes a discrete action path, in which the VLM autoregressively decodes action tokens for the entire action chunk. We measured this path under the same optimized graph as above. MOLMOACT2 runs at 14.17 Hz and MOLMOACT2-THINK at 6.82 Hz i.e., 3.94× and 1.86× slower than the corresponding continuous path. This gap comes from the large VLM decoding overhead, while the action expert emits the entire chunk in a small fixed number of flow matching steps through a smaller cross-attention head, with the VLM cache reused across steps. We therefore use the continuous path as the default deployment option.

7 Related work

Generalist robot manipulation policies. In recent years, robotic manipulation research has shifted focus from task-specific designs toward developing generalist policies capable of operating across diverse scenarios and embodiments (Berscheid et al., 2019; Brohan et al., 2022; Dasari et al., 2019; Ebert et al., 2021; Fang et al., 2023; Jang et al., 2022; Khazatsky et al., 2024; Mandlekar et al., 2018; Walke et al., 2023; Shafiuallah et al., 2023). A representative approach is the development of vision-language-action (VLA) models. In this paradigm, vision-language models (VLMs)—pretrained on vast, web-scale multimodal data (e.g., image, video, text)—are further trained on collection of diverse robotics datasets (Black et al., 2024; Brohan et al., 2022; Zheng et al., 2024; Zhao et al., 2025; Team et al., 2025b; Li et al., 2024b; Qu et al., 2025; Kim et al., 2024). This knowledge transfer significantly reduces the reliance on domain-specific robotics data, which is typically expensive to collect via teleoperation. However, with the notable exception of our previous work, MolmoAct (Lee et al., 2025), reproducing frontier VLA models (e.g., the π series (Black et al., 2024; Intelligence et al., 2025)) remains highly challenging due to closed-source training data and code. MOLMOACT represents our latest effort to democratize VLA research by releasing full training code, data, and model checkpoints that achieve state-of-the-art performance across a wide range of robotic manipulation benchmarks.

Embodied reasoning for robotic manipulation. While textual chain-of-thought (CoT) prompting (Wei et al., 2022) has advanced reasoning in domains like program synthesis and mathematics, its benefits for robotic manipulation remain limited, likely because tacit physical knowledge is rarely articulated in the text corpora used to train backbone LLMs. This has motivated a growing body of work on non-textual ‘embodied CoT’ that leverages visual or spatial representations, including object bounding boxes and points (Zawalski et al., 2024; Yuan et al., 2024; Li et al., 2025), future image predictions (Zhao et al., 2025), latent reasoning (Tur

Archaeologist · close

Molmo lineage || PI lineage — shared FAST→FM curriculum, different S2 tokens.

New: ER · YAM 720h · per-layer KV · adaptive depth · NOT HL / RTC / stage voting

Q&A · Oct 28 · MolmoAct2 / 2605.02881

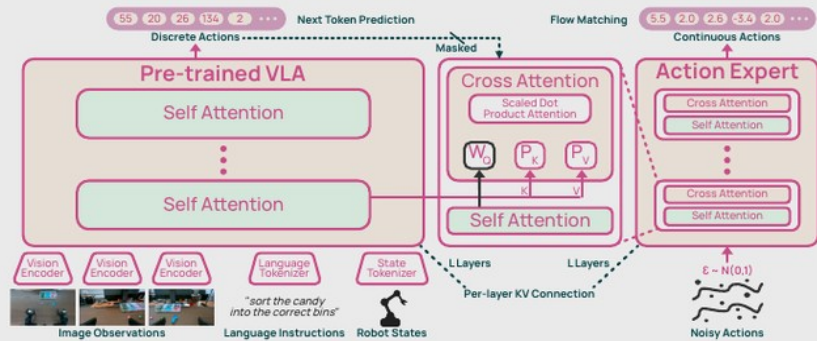


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Action and state representation. This stage is deliberately limited to discrete autoregressive supervision: the model predicts action tokens, but does not yet train the continuous action expert used in post-training. Each robot target is represented as a one-second action chunk, so the number of raw actions in the chunk follows the control frequency of the source dataset. Before tokenization, continuous action and state dimensions are normalized with 1-99 percentile statistics; gripper commands are treated separately from this percentile scaling when they are represented as binary or narrow-range open/close signals. Action vectors are padded to a 32-dimensional space and then encoded with the 2048-token vocabulary of MOLMOACT2-FAST TOKENIZER. Proprioceptive state is represented separately: after normalization, each state value is uniformly discretized into one of 256 state tokens and appended to the prompt before the action target.

Training. We train for 200K steps with a maximum sequence length of 4200 tokens. Examples vary substantially in length: a text-only example may use only a few hundred tokens, while multi-camera robot examples must carry images, states, and action targets. We therefore use on-the-fly packing to combine multiple short examples into one 4200-token sequence. Packed examples share the same forward pass, but their text, visual tokens, state tokens, and action targets remain separated by the training attention mask, so the model does not condition one example on another. Appendix B gives the implementation details for this packing procedure and the shared training stack. We train the vision encoder, connector, language model, and newly added tokens’ embeddings with a global batch size of 128 across 64 H100 GPUs for around 5,760 GPU hours. The vision encoder and connector use a learning rate of 5×10^{-6} , and the language model uses 1×10^{-5} . This produces a discrete VLA checkpoint that can later be adapted to continuous control.